20251209

ex $\det : GL_n(\mathbb{R}) \longrightarrow \mathbb{R}^\times , \quad A \longmapsto \det(A)$

$\nabla \det(A) = ?$

$$D\det(A)\cdot H = \underbrace{\det(A)}_{\in \mathbb{R}} \operatorname{tr}(A^{-1}H) = \operatorname{tr}\left((\det A)A^{-1}H\right)$$

$$= \left\langle \left(\det(A)A^{-1}\right)^T, H \right\rangle$$

$$\Rightarrow \nabla \det(A) = \det(A)A^{-T} = \underbrace{\operatorname{adj}(A)}_{\substack{\Downarrow \\ \text{adjugate matrix of } A}}{}^T$$

when A is invertible,

$$\operatorname{adj}(A) = \det(A)A^{-1}$$

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \qquad \text{define } C = \begin{pmatrix} C_{11} & \cdots & C_{1n} \\ \vdots & \ddots & \vdots \\ C_{n1} & \cdots & C_{nn} \end{pmatrix}$$

where

$$C_{ij} = \det\big(\overbrace{A \text{ w/ } i\text{-th row \& } j\text{-th column deleted}}^{(n-1)\times(n-1) \text{ matrix}}\big)$$

$$= \det \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ & a_{ij} & \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}_{(n-1)\times(n-1)}$$

and $\mathrm{adj}: M_n(\mathbb{R}) \to M_n(\mathbb{R})$

$$\begin{pmatrix} + & - & + & - \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

$$\mathrm{adj}\,A = \left((-1)^{i+j} C_{ji}\right) = \begin{pmatrix} +C_{11} & \cdots & \pm C_{n1} \\ - & \ddots & \vdots \\ \vdots & & \\ C_{1n} & \cdots & C_{nn} \end{pmatrix}$$

Consider $A \in M_3(\mathbb{R})$, $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$

$$\det(A) = a_{11}C_{11} - a_{12}C_{12} + a_{13}C_{13}$$

$$A(\mathrm{adj}\,A) = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}\begin{pmatrix} C_{11} & -C_{21} & C_{31} \\ -C_{12} & C_{22} & -C_{32} \\ C_{13} & -C_{23} & C_{33} \end{pmatrix}$$

$$= \begin{pmatrix} \det A & 0 & 0 \\ 0 & \det A & 0 \\ 0 & 0 & \det A \end{pmatrix} = (\mathrm{adj}\,A)\,A$$

$$23\text{-entry} = \overset{a_{31}}{\overset{\downarrow}{a_{21}}} C_{31} - \overset{a_{32}}{\overset{\downarrow}{a_{22}}} C_{32} + \overset{a_{33}}{\overset{\downarrow}{a_{23}}} C_{33}$$

$$= \det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ \underline{a_{21}} & \underline{a_{22}} & \underline{a_{23}} \\ \underline{a_{21}} & \underline{a_{22}} & \underline{a_{23}} \end{pmatrix} = 0$$

$$\left(A^{-1} = \frac{\mathrm{adj}\,A}{\det A} \right)$$

wikipedia, matrix calculus.

$$f(X) = \operatorname{tr}(AXBX^TC)$$

w/ X is a matrix, A, B, C of appropriate sizes.

$$Df(X)\cdot H = \frac{d}{dt}\Big|_{t=0} \operatorname{tr}\left(A(X+tH)B(X+tH)^TC\right)$$

$$= \cdots$$

or

$$Df(X)\cdot H = \operatorname{tr}(AH\boxed{BX^TC}) + \operatorname{tr}(AXBH^T\boxed{C})$$

$$= \operatorname{tr}(BX^TCAH) + \operatorname{tr}(CAXBH^T)$$

$$= \langle (BX^TCA)^T, H\rangle + \langle CAXB, H\rangle$$

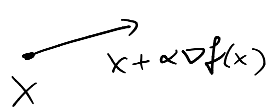
$$= \langle \underline{A^TC^TXB^T + CAXB}, H\rangle$$

$$\Rightarrow \nabla f(X) = A^TC^TXB^T + CAXB$$

$f : \mathbb{R}^n \to \mathbb{R}$, $x \mapsto x^TAx + b^Tx$

w/ $A \in M_n(\mathbb{R})$, $b \in \mathbb{R}^n$ const. parameters. fixed

$$\nabla f(x) = Ax + A^Tx + b \in \mathbb{R}^n$$



<u>ex</u> $f : M_n(\mathbb{R}) \longrightarrow \mathbb{R}$

$$A \longmapsto x^T A x \quad ,$$

$x \in \mathbb{R}^n$ constant vector fixed

$$Df(A) \cdot H = x^T H x = tr(x^T H x)$$

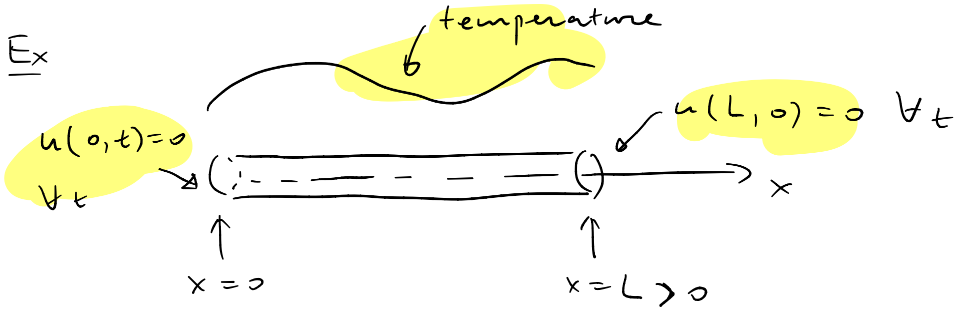
$$= tr(x x^T H) = \langle (x x^T)^T , H \rangle$$

$$\Rightarrow \quad \nabla f(A) = (x x^T)^T = x x^T$$

Sec 10.5 - 10.6 Boyce & Diprima.

Solve PDE, e.g., heat eqn

Ex



$$u : [0, L] \times [0, +\infty) \to \mathbb{R}$$

$$(x, t) \mapsto u(x,t)$$

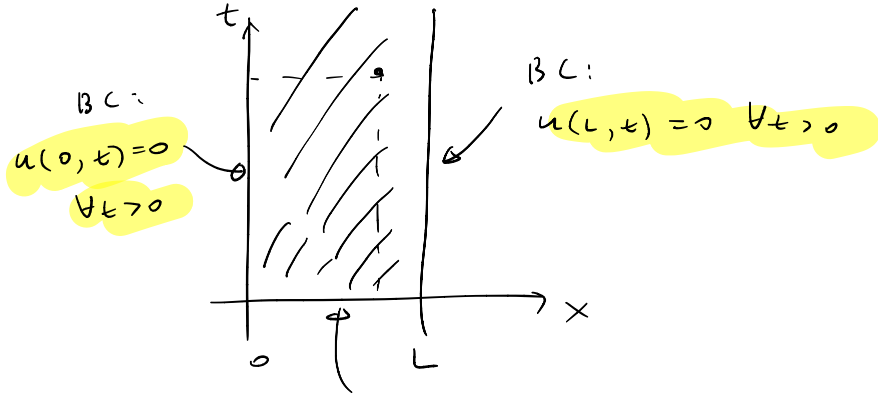
temperature of rod at position x & time t.

The time evolution of temperature u follows the heat eqn (heat conduction eqn)

$$\alpha^2 u_{xx} = u_t$$

w/ α^2 being a const, called thermal diffusivity

or
$$\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}, \quad t > 0$$



IC: $u(x,0) = f(x)\,, \quad 0 \le x \le L$

Separation of variables

We assume that $u(x,t) = X(x)\,T(t)$

where X & T are single variable function

then $\dfrac{\partial u}{\partial t} = \alpha^2 \dfrac{\partial^2 u}{\partial x^2}$ becomes

$$X(x)\,T'(t) = \alpha^2 X''(x)\,T(t)$$

which can be rearranged as

$$\frac{X''(x)}{X(x)} = \frac{1}{\alpha^2}\frac{T'(t)}{T(t)} =: -\lambda$$

then we get

$$\begin{cases} X'' + \lambda X = 0 \\ T' + \alpha^2 \lambda T = 0 \end{cases}$$

For the BC's

$$u(0,t) = X(0)\,T(t) = 0 \quad \forall t > 0$$

$$\Rightarrow X(0) = 0$$

Similarly, $X(L) = 0$

So for X, we get an eigenvalue problem

$$-X'' = \lambda X, \quad X(0) = X(L) = 0$$

The general sol'n to $X'' + \lambda X = 0$ is given by

① $\lambda < 0$, $\quad X(x) = c_1 e^{\sqrt{-\lambda}\,x} + c_2 e^{\sqrt{-\lambda}\,x}$,

if $X(0) = X(L) = 0$, then $c_1 = c_2 = 0$

② $\lambda = 0$, $\quad X(x) = c_1 x + c_2$

if $X(0) = X(L) = 0$, then $c_1 = c_2 = 0$

③ $\lambda > 0$, $\quad X(x) = c_1 \cos(\sqrt{\lambda}\,x) + c_2 \sin(\sqrt{\lambda}\,x)$

$$X(0) = c_1 \underbrace{\cos(\sqrt{\lambda}\,0)}_{=1} + \cancel{c_2 \sin(\sqrt{\lambda}\,0)} = 0 \Rightarrow c_1 = 0$$

$$X(L) = \cancel{c_1 \cos(\sqrt{\lambda}\,L)} + c_2 \underline{\underline{\sin(\sqrt{\lambda}\,L)}} = 0$$

$$\sin(\sqrt{\lambda}\,L) = 0 \Rightarrow \sqrt{\lambda}\,L = n\pi, \quad n = 1, 2, 3, \ldots$$

$$\Rightarrow \lambda_n = \left(\frac{n\pi}{L}\right)^2$$

$$\Rightarrow X_n(x) = \sin\left(\frac{n\pi x}{L}\right)$$

Back to T,

$$T' + \left(\frac{n^2 \pi^2 \alpha^2}{L^2}\right) T = 0 \,, \qquad n = 1, 2, 3, \ldots$$

$$\Rightarrow \quad T_n(t) = T(0)\, e^{-\frac{n^2 \pi^2 \alpha^2}{L^2} t}$$

So we see that each

$$u_n(x,t) = e^{-\frac{n^2 \pi^2 \alpha^2}{L^2} t} \sin\left(\frac{n\pi x}{L}\right) \,, \quad n = 1, 2, \ldots$$

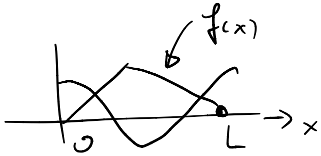
is a solution to the heat eqn.
(w/ BC already satisfied)

Assume that

$$u(x,t) = \sum_{n \geq 1} \underline{\underline{c_n}}\, e^{-\frac{n^2 \pi^2 \alpha^2}{L^2} t} \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = u(x, 0) = \sum_{n \geq 0} c_n \sin\left(\frac{n\pi x}{L}\right)$$

$$\int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx = \int_0^L \sin\left(\frac{n\pi x}{L}\right) \left(\sum_{n \geq 0} c_n \sin\left(\frac{n\pi x}{L}\right) \right) dx$$



$$= c_n \underbrace{\int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx}_{L/2}$$

$$\Rightarrow \quad c_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

FACT

$$\int_0^L \sin\frac{n\pi x}{L}\,\sin\frac{m\pi x}{L}\,dx = \begin{cases} 0, & m \neq n \\ \frac{n}{2}, & m = n \end{cases}$$